Quantum-Inspired vs. Classical Solvers: A Reproducible Benchmark

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Abstract

We present a reproducible benchmark comparing GQIR, SimCIM, SQA, SA, and CP-SAT on three QUBO problems across five sizes ($n = 20$–100). With 3,180 runs, we show SimCIM and SQA achieve near-zero gap on MaxCut, while classical SA dominates random QUBO (0.12% at $n = 100$). CP-SAT achieves 0% gap on MaxCut at all sizes. The QUBO penalty formulation for constrained problems is fundamentally ill-conditioned (gaps $> 10^3$% regardless of penalty scaling). All quantum-inspired results are from classical simulation; no quantum hardware was used.

1 Introduction

This paper benchmarks quantum-inspired algorithms (SimCIM, SQA, GQIR) against classical SA and CP-SAT on QUBO-formulated optimization problems.

2 Methods

All problems minimize $x^\top Qx$ for $x \in \{0,1\}^n$. **Random QUBO:** $Q_{ij} \sim \mathcal{U}[-1,1]$, density 0.3. **MaxCut:** $Q_{ii} = -\frac{1}{2}\sum_{j\neq i} w_{ij}$, $Q_{ij} = \frac{1}{2}w_{ij}$. **Knapsack:** Penalty formulation $Q_{ii} = -v_i + P(w_i^2 - 2Cw_i)$, $Q_{ij} = 2Pw_iw_j$.

Algorithms: GQIR (gradient-based relaxation), SimCIM (phase dynamics), SQA (path-integral MC), SA (Metropolis), CP-SAT (OR-Tools, 30 s timeout for $n > 20$).

3 Results

MaxCut

SimCIM achieves $< 10^{-14}$% gap. CP-SAT matches at 0%. SQA is < 0.1%. SA improves with n (0.64% $\rightarrow$ 0.03%). GQIR shows 3–7% gap.

Random QUBO

Classical SA dominates (0.12% at $n = 100$). CP-SAT is competitive (0.79–4.91%) but encoding overhead hurts at $n = 100$. SimCIM fails (> 130%). SQA improves with size (8% $\rightarrow$ 1%).

Penalty Pathology

All heuristics produce gaps $> 10^3$% on knapsack, regardless of penalty multiplier α. This is a fundamental limitation of QUBO penalty encoding.

4 Conclusion

(1) SimCIM/SQA excel on MaxCut. (2) SA dominates unstructured QUBO. (3) CP-SAT is an excellent reference for MaxCut. (4) QUBO penalty encoding is fundamentally ill-conditioned.

References

[1] S. Kirkpatrick et al., *Science*, 1983.

[2] T. Leleu et al., *Phys. Rev. A*, 2021.

[3] OpenJij Project, https://openjij.github.io/

[4] Google OR-Tools, https://developers.google.com/optimization/cp/cp_solver

[5] A. Lucas, *Front. Phys.*, 2014.

Table 1: Mean gap (%). CP-SAT† = 30 s timeout.

Algorithm	Problem	n	Gap(%)	Succ(%)	Time(s)
SimCIM	MaxCut	20	≈0	100	0.02
SimCIM	MaxCut	100	≈0	100	0.03
SQA	MaxCut	20	0.05	100	0.04
SQA	MaxCut	100	0.02	100	0.16
SA	MaxCut	20	0.64	74	0.01
SA	MaxCut	100	**0.03**	100	0.03
CP-SAT†	MaxCut	20	0.00	100	2.81
CP-SAT†	MaxCut	100	0.00	100	30.05
GQIR	MaxCut	20	3.39	0	0.04
GQIR	MaxCut	100	6.90	0	0.03
SA	RndQUBO	20	2.23	80	0.01
SA	RndQUBO	100	**0.12**	**98**	0.03
SQA	RndQUBO	20	7.98	22	0.04
SQA	RndQUBO	100	1.00	62	0.17
CP-SAT†	RndQUBO	20	4.91	70	0.03
CP-SAT†	RndQUBO	100	2.70	20	30.03
GQIR	RndQUBO	20	23.03	30	0.04
GQIR	RndQUBO	100	9.17	0	0.03

† Feasible under 30 s timeout.